authorizationManagement

Service Description

Service ID: *"authorizationManagement"*

Abstract
This document describes the `authorizationManagement` service, version 1.0.0, produced by the ConsumerAuthorization core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `authorizationManagement` service, which provides operator management of authorization policies at the management level, and bulk evaluation of authorization questions. It is produced by the ConsumerAuthorization system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Corresponds to the Arrowhead 4.x Authorization `mgmt` endpoints.

2.2 How This Service Is Meant to Be Used

Operators and configuration scripts grant and revoke management-level policies for any provider, query all policies, and check a list of consumer/provider/target triples. DynamicServiceOrchestration uses `check-policies` to filter candidate providers.

2.3 Important Delimitations and Dependencies

Access is governed by the management policy of ConsumerAuthorization; `check-policies` is also used by the systems in `management.whitelist`.

3 Service Interface

This section describes the interfaces to the `authorizationManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

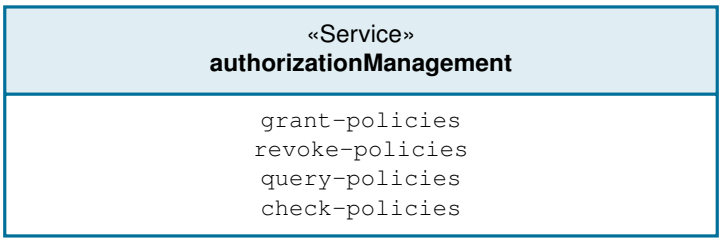


Figure 1: SysML-style block of the `authorizationManagement` service and its operations.

The following interface operations are available.

3.1 operation grant-policies (AuthorizationMgmtGrantListRequest) : AuthorizationPolicyListResponse

Creates management-level policies for the given providers and targets.

3.2 operation revoke-policies (List<Identifier>)

Removes policies by instance identifier.

3.3 operation query-policies (AuthorizationQueryRequest) : AuthorizationPolicyListResponse

Returns a page of policies of either level matching the filter.

3.4 operation check-policies (AuthorizationVerifyListRequest) : AuthorizationVerifyListResponse

Evaluates a list of authorization questions and returns the answer for each.

4 Information Model

Here, all data objects that can be part of the `authorizationManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.14, which are used to represent things like hashes and identifiers.

4.1 struct AuthorizationMgmtGrantListRequest

Management-level policies to create.

Field	Type	Description
list	List<AuthorizationMgmtGrantRequest>	The list of request items.

4.2 struct AuthorizationMgmtGrantRequest

A management-level authorization policy for one provider and target.

Field	Type	Description
cloud	Name	Cloud identifier of the consumer cloud the policy applies to; `LOCAL` for the own cloud.
provider	Name	Provider system the policy is set for.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
description	String	Free-text description.
defaultPolicy	AuthorizationPolicyRequest	Policy that applies to all operations without a scoped policy.
scopedPolicies	Map<String, AuthorizationPolicyRequest>	Policies for individual operations, keyed by operation name.

4.3 struct AuthorizationPolicy

One policy rule.

Field	Type	Description
policyType	AuthorizationPolicyType	Policy type: `ALL`, `WHITELIST`, `BLACKLIST` or `SYS_METADATA`.
policyList	List<String>	System names of the white- or blacklist.
policyMetadataRequirement	MetadataRequirement	Metadata the consumer system must have (`SYS_METADATA`).

4.4 struct AuthorizationPolicyListResponse

List of authorization policies.

Field	Type	Description
entries	List<AuthorizationPolicyResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.5 struct AuthorizationPolicyRequest

One policy rule.

Field	Type	Description
policyType	String	Policy type: `ALL`, `WHITELIST`, `BLACKLIST` or `SYS_METADATA`.
policyList	List<String>	System names of the white- or blacklist.
policyMetadata Requirement	MetadataRequirement	Metadata the consumer system must have (`SYS_METADATA`).

4.6 struct AuthorizationPolicyResponse

A stored authorization policy.

Field	Type	Description
instanceId	Identifier	Policy instance identifier `<MGMT or PR>|<cloud>|<provider>|<targetType>|<target>`.
level	AuthorizationLevel	Policy level: `MGMT` (set by an operator) or `PROVIDER` (set by the provider itself).
cloud	Name	Cloud identifier of the consumer cloud the policy applies to; `LOCAL` for the own cloud.
provider	Name	Provider system the policy protects.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
description	String	Free-text description.
defaultPolicy	AuthorizationPolicy	Policy that applies to all operations without a scoped policy.
scopedPolicies	Map<String, AuthorizationPolicy>	Policies for individual operations, keyed by operation name.
createdBy	Name	Name of the system that created the record.
createdAt	DateTime	UTC time of creation.

4.7 struct AuthorizationQueryRequest

Filter for authorization policies.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
level	String	Policy level to match, `MGMT` or `PROVIDER`.
providers	List<Name>	Provider system names to match.
instanceIds	List<Identifier>	Policy instance identifiers to match.
cloudIdentifiers	List<String>	Cloud identifiers to match.
targetNames	List<String>	Target names to match.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.

4.8 struct AuthorizationVerifyListRequest

Authorization checks to evaluate.

Field	Type	Description
list	List<AuthorizationVerifyRequest>	The list of request items.

4.9 struct AuthorizationVerifyListResponse

Results of authorization checks.

Field	Type	Description
entries	List<AuthorizationVerifyResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.10 struct AuthorizationVerifyRequest

One authorization question: may the consumer use the target (operation) of the provider?

Field	Type	Description
provider	Name	Provider system; for `verify` defaults to the requester.
consumer	Name	Consumer system; for `verify` defaults to the requester.
cloud	Name	Cloud of the consumer, `LOCAL` by default.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	Operation to check; empty checks the default policy.

4.11 struct AuthorizationVerifyResponse

The answer to one authorization question.

Field	Type	Description
provider	Name	Name of the provider system.
consumer	Name	Name of the consumer system.
cloud	Name	Cloud of the consumer.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	The checked operation.
granted	Boolean	Result of the check: true when the consumer is authorized.

4.12 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.13 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.14 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
AuthorizationLevel	One of `MGMT`, `PROVIDER`.
AuthorizationPolicyType	One of `ALL`, `WHITELIST`, `BLACKLIST`, `SYS_METADATA`.
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE`.
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, consumerauthorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-SysD.pdf.

[2] ——, “authorizationManagement HTTP/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationManagement-HTTP-IDD.pdf.

[3] ——, “authorizationManagement MQTT/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	